A SPEECHTECHJOBS GUIDE

The Speech AI Career Starter Kit

Everything here is condensed from our free articles at speechtechjobs.com/blog — follow the links for the full versions.

01 The speech-AI job map

Speech AI splits into a handful of specialties. Rough US total-comp ranges for mid-to-senior roles (see the salary guides for sourced detail and how to research a specific offer):

SPECIALTY	WHAT YOU WORK ON	ROUGH RANGE
ASR research	Novel models, SSL, publications	$200K–$350K+
ASR / speech ML engineering	Fine-tuning, eval, production ASR	$150K–$260K
Whisper specialist work	Domain adaptation, inference optimization	$140K–$220K
TTS / voice synthesis	Vocoders, codec-LMs, prosody	$150K–$260K
Speech analytics	Diarization + sentiment + summarization	$150K–$240K
Voice biometrics	Speaker verification, anti-spoofing	$165K–$230K
Voice agents	ASR+TTS+LLM pipelines, real-time	$150K–$250K
Embedded / on-device	Quantization, edge inference, DSP	$145K–$215K

Ranges vary widely by company stage, location, and level. Big labs and revenue-intelligence companies sit at the top; earlier-stage startups trade cash for equity.

02 How to break in

01 **Pick a track.** Research (novel models, publications), applied ML (fine-tuning, evaluation, data), or systems/inference (latency, cost, serving). The interview loops differ.

02 **Ship something with real audio.** A fine-tuned Whisper on a domain dataset, a streaming transcription demo, or a diarization + ASR pipeline with measured WER and DER beats a generic ML portfolio.

03 **Learn the metrics.** WER, CER, RTF (real-time factor), DER — and how to read an evaluation that isn't leaking training data.

04 **Contribute where the field lives.** Whisper, faster-whisper, NeMo, icefall, SpeechBrain, pyannote — a merged PR gets noticed.

05 **Do you need a PhD?** For pure research roles at top labs, usually yes — or an equivalent publication record. For applied and systems roles, no: a strong portfolio and one deep audio project is enough.

Full guide: speechtechjobs.com/blog/how-to-break-into-speech-tech-2026

03 The 10 ASR interview questions that come up most

01 **Explain WER. What are its limitations?** Insertions/deletions/substitutions over reference length; doesn't weight semantic importance; sensitive to normalization.

02 **CTC vs. attention vs. RNN-T — trade-offs?** Alignment assumptions, streaming capability, training stability.

03 **How does Whisper work, and where does it fail?** Encoder-decoder, chunked; hallucinates on silence, long-form drift, weak word-level timestamps.

04 **What is a WFST and why did classic ASR use it?** Composing HMM/lexicon/LM into one searchable graph; efficient decoding, lattices.

05 **How would you cut inference cost in half?** faster-whisper / CTranslate2, smaller or distilled model, batching, VAD, quantization.

06 **How do you build a fair evaluation set?** Held-out speakers and conditions, no leakage, representative of production audio.

07 **Streaming ASR — what changes?** Latency budget, partial hypotheses, endpointing, lookahead vs. accuracy.

08 **Speaker diarization pipeline — walk through it.** VAD → embeddings (x-vector / ECAPA) → clustering → resegmentation; DER; overlap handling.

09 **Fine-tuning: how do you know it helped?** WER/CER delta on a held-out domain set, not training loss; significance.

10 **When would you NOT use Whisper?** Real-time/streaming, strict word-level timestamps, or when a commercial API is simply cheaper.

Full set with answers: speechtechjobs.com/blog/speech-recognition-interview-questions-2026

04 The tools that matter

Whisper (OpenAI) — the default for "paste audio, get a good transcript." Strong multilingual accuracy out of the box; weak at streaming and precise timestamps. Optimize with faster-whisper / CTranslate2.

Classic Kaldi — the 2011 C++/WFST toolkit. Near-maintenance now, but still running in telephony, government, and enterprise systems. Knowing WFSTs, lattices and lexicons is a rare, transferable skill.

Next-gen Kaldi (k2 / lhotse / icefall / sherpa) — PyTorch-native, actively developed, and a leading choice for streaming and on-device ASR.

NVIDIA NeMo / Riva — production toolkit and serving stack; Parakeet and Canary model families.

wav2vec 2.0 / HuBERT / MMS (Meta) — self-supervised speech representations; the research lineage behind a lot of modern ASR.

Vosk — wraps Kaldi behind a simple offline API in ~20 languages; very widely used.

Full comparisons: speechtechjobs.com/blog/is-kaldi-still-used-2026 and /blog/kaldi-vs-whisper-vs-wav2vec-2026

05 Where the jobs are

DEDICATED ASR / SPEECH-TO-TEXT PLATFORMS

Deepgram · AssemblyAI · Speechmatics · Rev · Gladia · Otter.ai

BIG LABS & PLATFORMS

OpenAI · Google / DeepMind · Amazon (Alexa, Transcribe) · Apple · Microsoft (Azure Speech, Nuance) · Meta AI (FAIR) · NVIDIA

VOICE AGENTS / CONVERSATIONAL AI

Cartesia · Rime · PolyAI · Vapi · Retell AI · Cresta · Observe.AI · Gong

VERTICAL ASR

Suki · Abridge · DeepScribe · Nuance (healthcare) — Verbit · Rev (legal) — Cerence · SoundHound (automotive)

Full guide with careers links: speechtechjobs.com/blog/top-companies-hiring-asr-engineers-2026

06 The 2026 speech-conference calendar

CONFERENCE	DATES	LOCATION
ICASSP 2026	May 4–8, 2026	Barcelona, Spain (concluded)
Interspeech 2026	Sept 28 – Oct 1, 2026	Sydney, Australia
SLT 2026	Dec 13–16, 2026	Palermo, Italy

Note: there is no ASRU 2026 — ASRU runs in odd years (next: 2027). SLT is its even-year counterpart. ICASSP and Interspeech are the main recruiting venues; a paper gets you approached directly, but the hallway track works without one.

Details: speechtechjobs.com/blog/asru-2026 and /blog/speech-tech-conferences-2026

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